

Unit-2: Design websites using Bootstrap 4

2.1 Bootstrap introduction

→ What is Bootstrap?

Bootstrap is a free front-end framework for faster and easier web development. Bootstrap includes HTML and CSS based design templates for typography, forms, buttons, tables, navigation, modals, image carousels and many other, as well as optional JavaScript plug-ins. Bootstrap also gives you the ability to easily create responsive designs.

→ Responsive web design

Responsive web design is about creating web sites which automatically adjust themselves to look good on all devices, from small phones to large desktops of different screen size.

→ Bootstrap 4

Bootstrap 4 is the newest version of Bootstrap, which is the most popular HTML, CSS, and JavaScript framework for developing responsive, mobile-first websites. It is used for faster and easier web development. Bootstrap 4 is completely free to download and use.

→ Bootstrap 3 vs. Bootstrap 4

Bootstrap 4 is the newest version of Bootstrap; with new components, faster style sheet and more responsiveness.

Bootstrap 4 supports the latest, stable releases of all major browsers and platforms. However, Internet Explorer 9 and down is not supported.

→ **Advantages of Bootstrap:**

Easy to use: Anybody with just basic knowledge of HTML and CSS can start using Bootstrap

Responsive features: Bootstrap's responsive CSS adjusts to phones, tablets, and desktops.

Mobile-first approach: In Bootstrap, mobile-first styles are part of the core framework.

Browser compatibility: Bootstrap 4 is compatible with all modern browsers (Chrome, Firefox, Internet Explorer 10+, Edge, Safari, and Opera)

→ **There are two ways to use Bootstrap 4 on your own web site.**

1. Include Bootstrap 4 from a CDN (Content Delivery Network, e.g. MaxCDN)
2. Download Bootstrap 4 from getbootstrap.com

→ Bootstrap 4 general structure:

1. Add the HTML5 doctype

Bootstrap 4 uses HTML elements and CSS properties that require the HTML5 doctype. Always include the HTML5 doctype at the beginning of the page, along with the lang attribute and the correct character set:

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8">
  </head>
</html>
```

2. Bootstrap 4 is mobile-first

Bootstrap 4 is designed to be responsive to mobile devices. Mobile-first styles are part of the core framework.

To ensure proper rendering and touch zooming, add the following <meta> tag inside the <head> element:

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

The width=device-width part sets the width of the page to follow the screen-width of the device (which will vary depending on the device).

The initial-scale=1 part sets the initial zoom level when the page is first loaded by the browser.

3. Containers

Bootstrap 4 also requires a containing element to wrap site contents.

There are two container classes to choose from:

1. The .container class: provides a responsive fixed width container.

2. The .container-fluid class: provides a full width container, spanning the entire width of the viewport.

→ **Example**

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap 4 Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.mi
n.css">
</head>
<body>

<div class="container">
  <h1>My First Bootstrap Page</h1>
  <p>This is some text.</p>
</div>

</body>
</html>
```

→ Bootstrap Grid structure

Bootstrap's grid system is built with flex box and allows up to 12 columns across the page.

If you do not want to use all 12 columns individually, you can group the columns together to create wider columns.

span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1
span 4				span 4				span 4			
span 4				span 8							
span 6						span 6					
span 12											

The grid system is responsive, and the columns will re-arrange automatically depending on the screen size.

Make sure that the sum adds up to 12 or fewer (it is not required that you use all 12 available columns).

→ Container classes

Bootstrap 4 uses container classes to wrap the page's contents. It contains two container classes –

.container – It represents a fixed width container.

.container-fluid – It represents a full width container.

1) Container

The .container class is used to wrap the page content with fixed width and content can be placed in the center easily by using the .container class as shown below.

```
<div class = "container">
```

```
...
```

```
</div>
```

2) Fluid Container

You can create a full width container by using the .container-fluid class as shown below.

```
<div class = "container-fluid">
```

```
...  
</div>
```

2.2 Grid Structure (Grid Classes)

The Bootstrap 4 grid system has five classes:

- .col- (extra small devices - screen width less than 576px)
- .col-sm- (small devices - screen width equal to or greater than 576px)
- .col-md- (medium devices - screen width equal to or greater than 768px)
- .col-lg- (large devices - screen width equal to or greater than 992px)
- .col-xl- (xlarge devices - screen width equal to or greater than 1200px)

The classes above can be combined to create more dynamic and flexible layouts.

Note: Each class scales up, that means sm class scales up from sm to xl. So it supports the largest screen size xl to sm.

The following example shows how to create two equal-width columns, on all devices and screen widths:

```
<div class="row">  
  <div class="col">ONE </div>  
  <div class="col">TWO</div>  
</div>
```

The syntax to specify bootstrap grid class is:

class = "col-gridClass-spanValue"

gridClass = sm / md / lg / xl / * (small, medium, large, extra Large, all)
spanValue = between 1 to 12

Example: will create 4 equal width columns, responsive up to small devices.

```
<div class="row">  
  <div class="col-sm-3">.col-sm-3</div>  
  <div class="col-sm-3">.col-sm-3</div>  
  <div class="col-sm-3">.col-sm-3</div>  
  <div class="col-sm-3">.col-sm-3</div>  
</div>
```

Note: the total of span value must be 12, as one row is divided into 12 spans.

Example: will create 2 unequal responsive columns

```
<div class="row">
  <div class="col-sm-4">.col-sm-4</div>
  <div class="col-sm-8">.col-sm-8</div>
</div>
```

→ Working of Grid System

- Rows must be placed within a .container class for proper alignment and padding.
- For responsive width use .container class, and for fixed width across all viewport use the .container-fluid class.
- Use rows to create horizontal groups of columns.
- Content should be placed within the columns, and only columns may be the immediate children of rows.
- Columns contain padding for controlling the space between them.
- If you place more than 12 columns in a row, then the columns will be placed in a new line.
- Columns create gaps between column content via padding. Therefore, you can remove the margin from rows and padding from columns with .no-gutters class on the row.
- You can make grid system responsive by using five grid breakpoints such as extra small, small, medium, large, and extra large.
- Predefined grid classes like .col-4 are available for quickly making grid layouts.

2.3 Table, Colors, Alerts, Form Controls

→ Bootstrap 4 Colors

Text Colors

Bootstrap 4 has some contextual classes that can be used to provide "meaning colors".

The classes for text colors are: `.text-muted`, `.text-primary`, `.text-success`, `.text-info`, `.text-warning`, `.text-danger`, `.text-secondary`, `.text-white`, `.text-dark`, `.text-body` (default body color/often black) and `.text-light`.

This text is muted.

This text is important.

This text indicates success.

This text represents some information.

This text represents a warning.

This text represents danger.

Secondary text.

Dark grey text.

Body text.

Note: Contextual text classes can also be used on links, which will add a darker hover color.

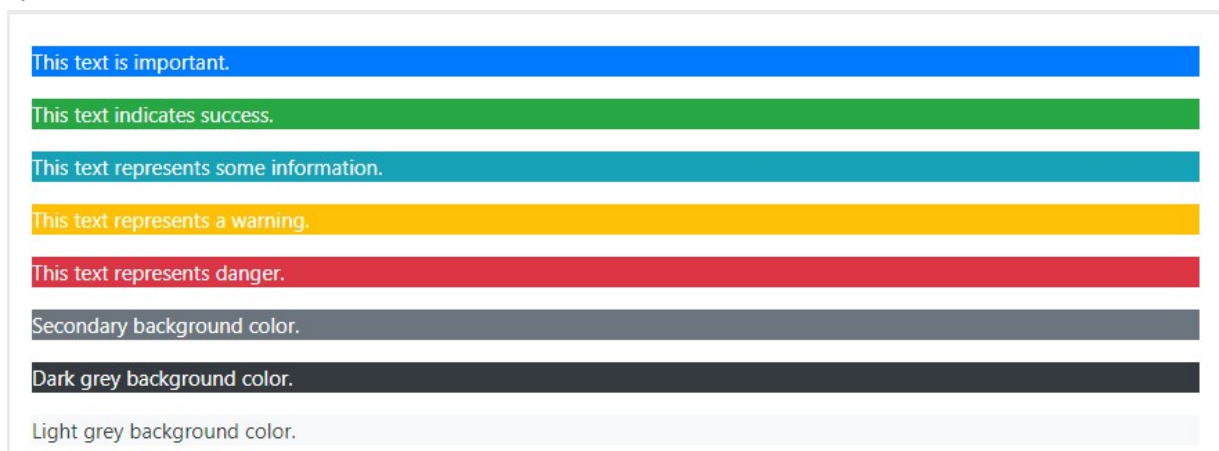
Background Colors

The classes for background colors are: `.bg-primary`, `.bg-success`, `.bg-info`, `.bg-warning`, `.bg-danger`, `.bg-secondary`, `.bg-dark` and `.bg-light`.

Note that background colors do not set the text color, so in some cases you'll want to use them together with a `.text-*` class.

Example code for background and text colors:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.mi
n.css">
</head>
<body>
<div class="container">
  <h2>Contextual Backgrounds</h2>
  <p>Note that you can also add a .text-* class if you want a
different text color:</p>
  <p class="bg-primary text-white">This text is important.</p>
  <p class="bg-success text-white">This text indicates success.</p>
  <p class="bg-info text-white"> represents some information.</p>
  <p class="bg-warning text-white">This text represents a warning.</p>
  <p class="bg-danger text-white">This text represents danger.</p>
  <p class="bg-secondary text-white">Secondary background color.</p>
  <p class="bg-dark text-white">Dark grey background color.</p>
  <p class="bg-light text-dark">Light grey background color.</p>
</div>
</body>
</html>
```



→ Bootstrap 4 Tables

- A basic Bootstrap 4 table has a light padding and horizontal dividers. The **.table** class adds basic styling to a table.
- The **.table-striped** class adds zebra-stripes to a table.
- The **.table-bordered** class adds borders on all sides of the table and cells.
- The **.table-hover** class adds a hover effect (grey background color) on table rows.
- The **.table-dark** class adds a black background to the table.
- Combine **.table-dark** and **.table-striped** to create a dark, striped table.
- The **.table-borderless** class removes borders from the table.
- The **.table-sm** class makes the table smaller by cutting cell padding in half.
- The **.table-responsive** class adds a scrollbar to the table when needed (when it is too big horizontally)

Contextual classes can be used to color the whole table (<table>), the table rows (<tr>) or table cells (<td>).

Class	Description
<code>.table-primary</code>	Blue: Indicates an important action
<code>.table-success</code>	Green: Indicates a successful or positive action
<code>.table-danger</code>	Red: Indicates a dangerous or potentially negative action
<code>.table-info</code>	Light blue: Indicates a neutral informative change or action
<code>.table-warning</code>	Orange: Indicates a warning that might need attention
<code>.table-active</code>	Grey: Applies the hover color to the table row or table cell
<code>.table-secondary</code>	Grey: Indicates a slightly less important action
<code>.table-light</code>	Light grey table or table row background
<code>.table-dark</code>	Dark grey table or table row background

→ Bootstrap 4 Alerts

Bootstrap 4 provides an easy way to create predefined alert messages. Alerts are created with the `.alert` class, followed by one of the contextual classes **`.alert-success`, `.alert-info`, `.alert-warning`, `.alert-danger`, `.alert-primary`, `.alert-secondary`, `.alert-light` or `.alert-dark`.**

```
<body>
<div class="container">
  <h2>Alerts</h2>
  <p>Alerts are created with the .alert class, followed by a
contextual color classes:</p>
  <div class="alert alert-success">
    <strong>Success!</strong> This alert box could indicate a
successful or positive action.
  </div>
</body>
```

O/P:

Alerts

Alerts are created with the `.alert` class, followed by a contextual color classes:

Success! This alert box could indicate a successful or positive action.

→ Bootstrap 4 Form Controls

Form controls automatically receive some global styling with Bootstrap. All textual `<input>`, `<textarea>`, and `<select>` elements with class **`.form-control`** have a width of 100%.

Bootstrap provides two types of form layouts:

- Stacked (full-width) form (by adding `.form-control` class to `<input>` elements)
- Inline form (by adding `.form-inline` class to `<form>` element)

The following **example** creates a **stacked form** with two input fields, one checkbox, and a submit button.

Add a wrapper element with **`.form-group`**, around each form control, to ensure proper margins. (optional)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.mi
n.css">
</head>
<body>
<div class="container">
  <h2>Stacked form</h2>
<form action="/action_page.php">
  <div class="form-group">
    <label for="email">Email address:</label>
```

```
<input type="email" class="form-control" placeholder="Enter email"
id="email">
</div>
<div class="form-group">
  <label for="pwd">Password:</label>
  <input type="password" class="form-control" placeholder="Enter
password" id="pwd">
</div>
<div class="form-group form-check">
  <label class="form-check-label">
    <input class="form-check-input" type="checkbox"> Remember me
  </label>
</div>
<button type="submit" class="btn btn-primary">Submit</button>
</form>
```

O/P:

Stacked form

Email:

Password:

☐ Remember me

In an **inline form**, all of the elements are inline and left-aligned.

*Additional rule for an inline form is that we have to add **class .form-inline** to the <form> element.*

Inline form with two input fields, one checkbox, and one submit button looks like following.

```
<form class="form-inline" action="/action_page.php">
<label for="email">Email address:</label>
```

```
<input type="email" class="form-control" placeholder="Enter email"
id="email">
<label for="pwd">Password:</label>
<input type="password" class="form-control" placeholder="Enter
password" id="pwd">
<div class="form-check">
  <label class="form-check-label">
    <input class="form-check-input" type="checkbox"> Remember me
  </label>
</div>
<button type="submit" class="btn btn-primary">Submit </button>
</form>
```

Bootstrap Inline Form

Email: Password: ☐ Remember me

Form with Bootstrap grid

You can also use columns (.col) to control the width and alignment of form inputs without using spacing utilities. Just remember to put them inside a .row container. If you want less grid margins (override default column gutters), use .form-row instead of .row.

Form Validations

You can use different validation classes to provide valuable feedback to users. Add either **.was-validated** or **.needs-validation** to the <form> element, depending on whether you want to provide validation feedback before or after submitting the form. The input fields will have a green (valid) or red (invalid) border to indicate what's missing in the form. You can also add a **.valid-feedback** or **.invalid-feedback** message to tell the user explicitly what's missing, or needs to be done before submitting the form.

```
<form action="/action_page.php" class="was-validated">
  <div class="form-group">
    <label for="uname">Username:</label>
    <input type="text" class="form-control" id="uname"
      placeholder="Enter username" name="uname" required>
    <div class="valid-feedback">Valid.</div>
```

```
    <div class="invalid-feedback">Please fill out this field.</div>
  </div>
  ...
</form>
```

2.4 Buttons and Button Groups

→ Buttons

Button Styles: Bootstrap 4 provides different styles of buttons.

```
<button type="button" class="btn">Basic</button>
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-secondary">Secondary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-dark">Dark</button>
<button type="button" class="btn btn-light">Light</button>
<button type="button" class="btn btn-link">Link</button>
```



Note: The button classes can be used on `<a>`, `<button>`, or `<input>` elements.

→ **Button Outline:** Bootstrap 4 provides eight outline/bordered buttons.

```
<button type="button" class="btn btn-outline-primary">Primary</button>
<button type="button" class="btn btn-outline-secondary"> Secondary
</button>
<button type="button" class="btn btn-outline-success">Success</button>
<button type="button" class="btn btn-outline-info">Info</button>
<button type="button" class="btn btn-outline-warning">Warning</button>
<button type="button" class="btn btn-outline-danger">Danger</button>
<button type="button" class="btn btn-outline-dark">Dark</button>
<button type="button" class="btn btn-outline-light text-dark"> Light
</button>
```



Button Sizes: We can use the **.btn-lg** class for large buttons or **.btn-sm** class for small buttons.

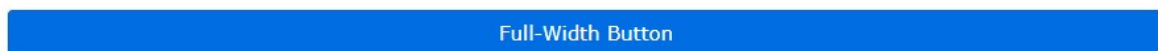
```
<button type="button" class="btn btn-primary btn-lg">Large</button>
```

```
<button type="button" class="btn btn-primary">Default</button>  
<button type="button" class="btn btn-primary btn-sm">Small</button>
```



Block Level Buttons: Add class **.btn-block** to create a block level button that spans the entire width of the parent element.

```
<button type="button" class="btn btn-primary btn-block"> Full-Width  
Button </button>
```



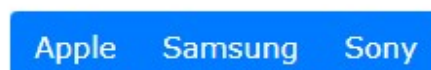
Active/Disabled Buttons: A button can be set to an active (appear pressed) or a disabled (unclickable) state. The class **.active** makes a button appear pressed, and the disabled attribute makes a button unclickable.

```
<button type="button" class="btn btn-primary" active> Active Primary  
</button>  
<button type="button" class="btn btn-primary" disabled> Disabled  
Primary </button>
```

→ Button Groups

Bootstrap 4 allows you to group a series of buttons together (on a single line) in a button group. Use a `<div>` element with class **.btn-group** to create a button group.

```
<div class="btn-group">  
  <button type="button" class="btn btn-primary">Apple</button>  
  <button type="button" class="btn btn-primary">Samsung</button>  
  <button type="button" class="btn btn-primary">Sony</button>  
</div>
```



Note: Instead of applying button sizes to every button in a group, use class **.btn-group-lg** for a large button group or the **.btn-group-sm** for a small button group.

Bootstrap 4 also supports **vertical button groups**. Use the class **.btn-group-vertical** to create a vertical button group.

2.5 Images, media objects

Bootstrap image shape classes

1) The **.rounded** class adds rounded corners to an image.

```

```

2) The **.rounded-circle** class shapes the image to a circle.

```

```

3) The **.img-thumbnail** class shapes the image to a bordered thumbnail.

```

```

Aligning images

→ Float an image to the right with the **.float-right** class or to the left with **.float-left**.

```

```

```

```

→ Center an image by adding the utility classes **.mx-auto (margin:auto)** and **.d-block (display:block)** to the image.

```

```

Responsive images

Responsive images automatically adjust to fit the size of the screen. Create responsive images by adding an **.img-fluid** class to the `<img>` tag. The image will then scale nicely to the parent element.

```

```

Bootstrap Media Objects

→ Bootstrap's media objects are used to construct highly repetitive components like blog comments, tweets, and the like. Bootstrap provides an easy way to align media objects (like images or videos) to the left or to the right of some content. This can be used to display blog comments, tweets and so on.

→ Use a `<div>` element with the **.media** class to create a container for media objects.

→ Use the **.media-left** class to align the media object (image) to the left, or the **.media-right** class to align it to the right.

→ Text that should appear next to the image, is placed inside a container with **class="media-body"**.

→ Additionally, you can use **.media-heading** for headings.

```
<div class="media">
  <div class="media-left">
    
  </div>
  <div class="media-body">
    <h4 class="media-heading">John Doe</h4>
    <p>Lorem ipsum...</p>
  </div>
</div>
```



John Doe

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor dolore magna aliqua.

→ The media object can also be top, middle or bottom aligned with the **media-top, media-middle or media-bottom** classes.

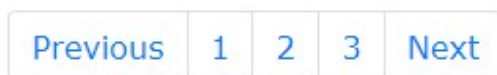
→ Media objects can also be nested (a media object inside a media object).

2.6 Bootstrap Pagination

1) Basic Pagination

If you have a web site with lots of pages, you may wish to add some sort of pagination to each page. To create a basic pagination, add the **.pagination** class to an `<ul>` element. Then add the **.page-item** to each `<li>` element and a **.page-link** class to each link inside `<li>`.

```
<ul class="pagination">
  <li class="page-item"><a class="page-link" href="#">Previous</a></li>
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item"><a class="page-link" href="#">2</a></li>
  <li class="page-item"><a class="page-link" href="#">3</a></li>
  <li class="page-item"><a class="page-link" href="#">Next</a></li>
</ul>
```



→ The **.active** class is used to "highlight" the current page.

```
<ul class="pagination">
  <li class="page-item"><a class="page-link" href="#">Previous</a></li>
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item active"><a class="page-link" href="#">2</a></li>
  <li class="page-item"><a class="page-link" href="#">3</a></li>
  <li class="page-item"><a class="page-link" href="#">Next</a></li>
</ul>
```



→ Add class **.disabled** to `<li>` element if a link for some reason is disabled. A disabled link cannot be clicked.

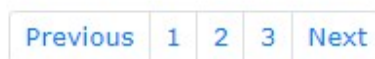


→ Pagination blocks can also be sized to a larger size or a smaller size. Add class **.pagination-lg** for larger blocks or **.pagination-sm** for smaller blocks.

```
<ul class="pagination pagination-lg">
  <li class="page-item"><a class="page-link" href=""> Previous</a></li>
```

```
<li class="page-item"><a class="page-link" href="#">1</a></li>
<li class="page-item"><a class="page-link" href="#">2</a></li>
<li class="page-item"><a class="page-link" href="#">3</a></li>
<li class="page-item"><a class="page-link" href="#">Next</a></li>
</ul>
```

```
<ul class="pagination pagination-sm">
  <li class="page-item"><a class="page-link" href="#">Previous</a></li>
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item"><a class="page-link" href="#">2</a></li>
  <li class="page-item"><a class="page-link" href="#">3</a></li>
  <li class="page-item"><a class="page-link" href="#">Next</a></li>
</ul>
```



→ Another form for pagination, is breadcrumbs. The **.breadcrumb** class indicates the current page's location within a navigational hierarchy. The **.breadcrumb** and **.breadcrumb-item** classes indicate the current page's location within a navigational hierarchy.

```
<ul class="breadcrumb">
  <li class="breadcrumb-item"><a href="#">Photos</a></li>
  <li class="breadcrumb-item"><a href="#">Summer 2017</a></li>
  <li class="breadcrumb-item"><a href="#">Italy</a></li>
  <li class="breadcrumb-item active">Rome</li>
</ul>
```

Photos / Summer 2017 / Italy / Rome

2.8 Bootstrap Themes (Templates)

A theme here means a readymade designed HTML – CSS template that a web designer can use for his website. A designer may modify the template as per his need and creativity.

There are no free readymade themes or templates provided by Bootstrap. But designer can use bootstrap grid structure, navigation bar and other bootstrap classes to design different themes or templates for their website.

But there is a bootstrap theme store (<https://themes.getbootstrap.com/>), where designers can purchase these readymade themes. Themes are built on Bootstrap as their own extended frameworks, rich with new components and plugins, documentation, and powerful build tools.

There are some other websites that provide free bootstrap templates, Like bootstrapmade.com and startbootstrap.com.